



Savitribai Phule Pune University
Centre For Distance and Online Education

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CENTRE FOR DISTANCE AND ONLINE EDUCATION

Program	Master of Computer Application	
Course	MJ Programming from First Principles	
Topic Name	standard data types - Introduction to standard data types	
Topic Code	-	
Unit/Module No. & Name	2	Standard Data Types
Self-Learning Hours	-	

Topic Details

S.No	Topic	Pg.No
1.0	Introduction	5
2.0	Meaning And Definition	6-9
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OBJECTIVE

1. A computer program works with data at every step.
2. Data can be a number, a letter, a word, or a true-or-false value.
3. A data type is a rule that tells the computer what a value means.
4. A data type also tells how much memory space the value needs.
5. It defines which actions or operations are allowed on that value.
6. The objective of this paper is to explain standard data types in clear and simple English.
7. The explanation keeps an academic tone suitable for undergraduate students.
8. The focus is on how data types help programs store data correctly.
9. Data types also help programs perform calculations safely.
10. They help compare values correctly and avoid common errors such as overflow and incorrect rounding.
11. After reading, you should be able to define common data types such as integer, floating-point, character, Boolean, and string.
12. You should be able to explain how data types are related to memory.
13. You should understand why data types have fixed sizes and limits.
14. You should be able to explain how choosing a data type can change program results.
15. You should be able to select a suitable data type for a simple task.
16. You should be able to explain your choice clearly in academic writing.
17. You should be able to identify problems that occur when a data type is used incorrectly.
18. In undergraduate studies, data types are taught early in the course.

19. This is because data types support later topics such as data structures and algorithms.
20. They are also important for databases and computer architecture.
21. When you write code for laboratory work, data types help show that your solution is correct.
22. When you write reports, data types help explain why a program behaves in a certain way.
23. In research-style writing, clearly stating the type system improves understanding.
24. It helps readers understand the limits of stored values.
25. It also helps readers understand the expected accuracy of computations.



1.0 INTRODUCTION

Computers do not guess what a value means. They store everything as bits, which are 0s and 1s. A data type gives meaning to these bits. For example, the same group of bits can represent a whole number, a decimal number, or a letter, depending on the type used. If the wrong type is chosen, the computer may read the bits incorrectly and give wrong results. In many systems, common data types are shared across different programming languages, even though their names or sizes may differ. This is why learning standard data types helps you work with many programming languages.

1.1 Data As Bits And Memory

A computer stores data in memory using bits. Bits are grouped into bytes, and bytes are combined to store larger values. In most systems, one byte contains 8 bits. A data type tells the computer how many bytes are needed for a value and how to understand the bits stored. For example, an integer may use 4 bytes, while a floating-point number may use 8 bytes. A character may use 1 byte in some systems and more bytes in others. Because of this, choosing the right data type affects both accuracy and memory usage.

1.2 Why Types Matter In Programs

Data types are important because they help prevent errors. They allow the compiler or interpreter to detect mistakes early, such as trying to add a number and a word together. Data types also help programs run faster and more efficiently, because the computer knows which hardware operation to use. For example, integer calculations and decimal calculations use different parts of the processor. When types are clearly defined, programs behave more predictably.

1.3 Standard Types As Shared Building Blocks

Standard data types are basic elements provided by most programming languages. They are used to create more complex data structures such as arrays, records, classes, and files. Even modern language features depend on these basic types. For example, a list of ages still uses integers, and a text message still uses characters or strings. Understanding standard data types makes it easier to understand how complex data is stored and processed.

2.0 MEANING AND DEFINITION

A standard data type is a commonly used type that appears in many programming languages and computer systems. It defines (1) the kind of value stored, such as number or text, (2) the range of valid values, such as minimum and maximum, (3) how the value is represented in memory, and (4) the basic operations allowed. Standard types are important because they allow programs to be written with predictable behavior.

2.1 Integer

An integer stores whole numbers such as -7 , 0 , and 42 . Integers do not store a decimal point. They are used for counting, indexing, and exact math where fractions are not needed. Integers have limits because memory space is finite. For example, if an integer uses a fixed number of bits, it can store only a certain range. If a program tries to store a value outside that range, the result may be an error or an incorrect wraparound value, depending on the language and system.

2.1.1 Signed And Unsigned Integers

A signed integer can represent both positive and negative numbers. An unsigned integer represents only nonnegative numbers. Unsigned types often allow a larger maximum value when the same number of bits is used, because no bits are used for the sign. Signed integers are common for general arithmetic. Unsigned integers are common for sizes, counts that cannot be negative, and bit-level operations.

2.2 Floating-Point

A floating-point type stores numbers with fractions, such as 3.14 or -0.25 . Floating-point is used for measurements, scientific values, averages, and many real-world quantities. Floating-point values are often approximations, not perfectly exact. This happens because many decimals cannot be stored exactly in binary form. As a result, small rounding differences can appear, and they can matter in some tasks. Good academic practice is to state when a result is approximate and to justify the chosen precision.

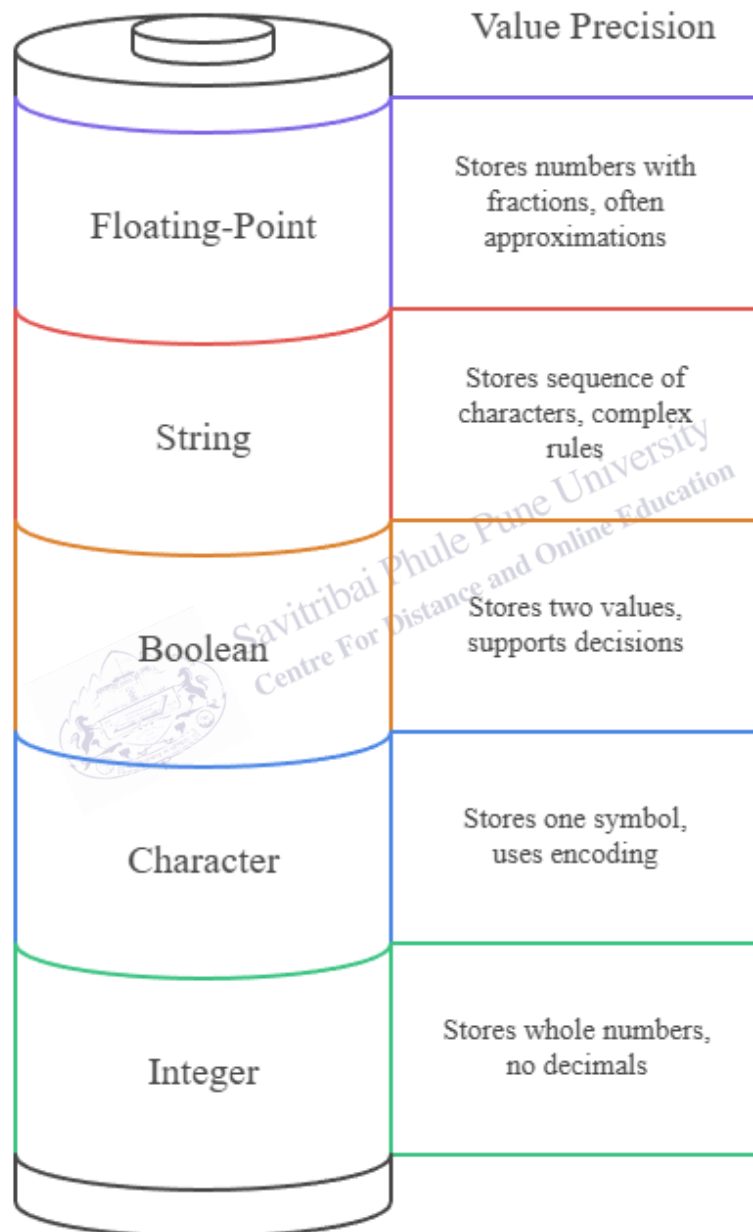
2.2.1 Precision And Rounding

Floating-point uses a limited number of bits, so it cannot store all real numbers exactly. It rounds to the nearest value that fits. In long calculations, small rounding differences can add up. For example, adding many small values may create a final result that is slightly different from the expected mathematical result. This does not mean floating-point is “wrong.” It means

floating-point is a model with limits. A careful programmer selects suitable precision and tests for acceptable error ranges.

Fig. 1

Data types range from exact to approximate values.



Standard data types move from exact to approximate values. Integers store whole numbers. Characters store single symbols. Booleans store true or false. Strings store sequences of characters. Floating-point stores fractional numbers with limited precision and possible approximations in computing systems.

2.3 Character

A character type stores one symbol, such as ‘A’, ‘7’, or ‘?’. Characters are often stored using codes. A common coding idea is that each symbol maps to a numeric value. Older systems often used ASCII for basic English symbols. Many modern systems use Unicode so that many world languages can be represented. In academic writing, it is useful to note that character storage depends on encoding, and encoding affects how much memory a character needs.

2.3.1 Encoding And Text Meaning

Encoding matters because a program may read text incorrectly if the wrong encoding is assumed. For example, the same byte values can represent different symbols under different rules. Unicode encodings like UTF-8 can use different numbers of bytes for different characters. This improves language support but adds complexity. When working with files and networks, encoding should be stated clearly.

2.4 Boolean

A Boolean type stores two possible values, usually written as true and false. Booleans support decision making in programs. They are used in conditions like “if,” “while,” and logical comparisons. Booleans help programs be clear because they separate facts from numbers. For example, “isLoggedIn” is clearer as a Boolean than as an integer, even if the integer uses 0 and 1. In undergraduate work, clear Boolean naming improves both code readability and report quality.

2.4.1 Logical Operators

Boolean values combine using logical operators. AND is true only if both parts are true. OR is true if at least one part is true. NOT flips the value. These operators support many real tasks like checking whether a user has permission or whether a sensor value is within a safe range.

2.5 String

A string is a sequence of characters used to represent text, like names, sentences, and messages. Strings are common in user interfaces, files, and web systems. In many languages, strings are more complex than characters because they include length rules, indexing rules, and memory rules. Some languages treat strings as arrays of characters. Others treat strings as special objects. Strings also raise questions about encoding, because text can include many symbols beyond English letters.

2.5.1 Length, Indexing, And Safety

Strings can cause errors when lengths are ignored. For example, writing beyond the end of a string buffer can cause security problems in low-level languages. Safe handling includes checking length, using correct library functions, and avoiding unsafe copying methods. In academic work, mentioning safe string handling shows awareness of secure programming practices.

2.6 Type Declaration, Type Inference, And Casting

Some languages require explicit type declarations. Others can infer types from context. In both cases, type rules still exist. Type casting is converting a value from one type to another, such as from integer to float. Casting can be safe or risky. For example, converting a float to an integer may lose the fractional part. Converting a large integer to a smaller integer type may lose high bits. Clear programs use casting carefully and explain it in comments or reports when it affects results.



3.0 NATURE OR TYPE

Standard data types can be grouped in more than one way. One common way is to split types into primitive types and non-primitive types. Primitive types include basic numbers, characters, and Booleans. Non-primitive types include arrays, records, and other constructed forms. Another grouping focuses on numeric behavior, such as exact integers versus approximate floating-point. Another grouping focuses on the typing system of a language, such as static typing versus dynamic typing. These categories help you compare languages and predict how errors will appear.

3.1 Primitive And Non-Primitive Types

Primitive types are the simplest. They are often supported directly by hardware and core language rules. Non-primitive types are built from primitive types. For example, an array of integers is non-primitive, but it is built from many integer elements. A record may store a mix of integers, Booleans, and strings. This view is helpful because many performance and memory effects depend on whether the type is simple or constructed.

3.1.1 Derived Types

Derived types include arrays, pointers, and sometimes references. They depend on primitive types because they store primitive values or addresses. Derived types matter because they can change memory layout and access patterns. For example, an array stores values in a contiguous block of memory, which can be efficient. A linked list stores nodes in separate memory locations, which can be less efficient due to pointer chasing.

3.1.2 User-Defined Types

User-defined types include structures, classes, and records. They allow a programmer to model real objects, such as a student record with an ID, name, and grade. User-defined types improve organization and readability. They also help ensure that related values stay together and are handled consistently. In academic reports, user-defined types are often justified as a way to match the problem domain clearly.

3.2 Numeric Categories

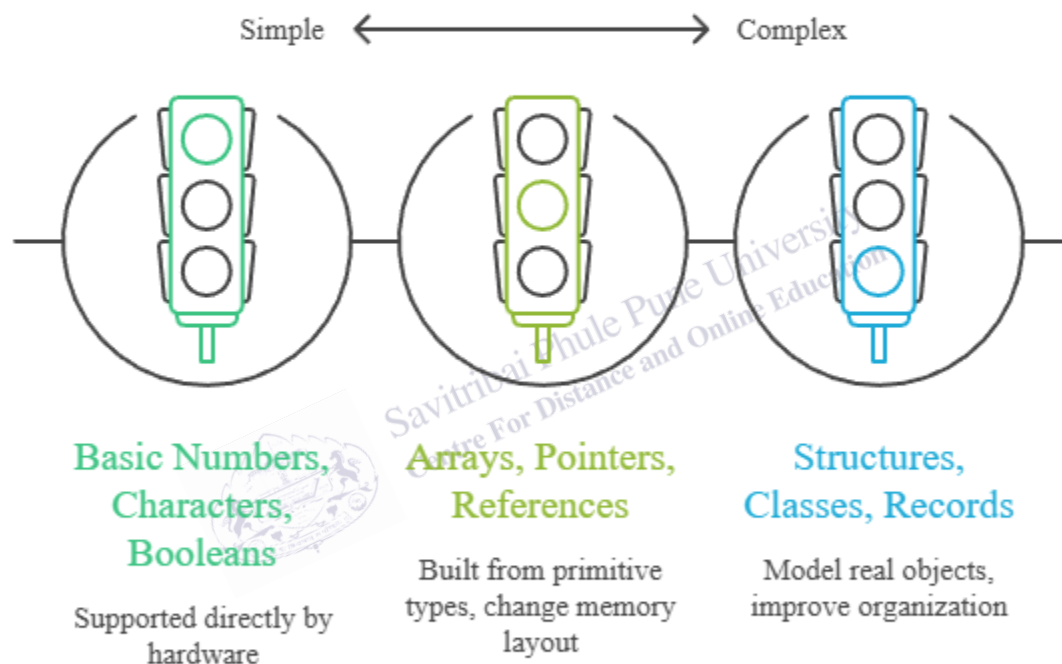
Numeric types can be exact or approximate. Integers are exact within their range. Floating-point numbers are approximate for many decimal fractions. Numeric categories also include size and sign choices. A larger numeric type can store a wider range but may use more memory. A smaller numeric type can save memory but may overflow more easily.

3.2.1 Range And Overflow

Each numeric type has a range. When a program tries to store a value outside that range, overflow can occur. Some languages raise errors, while others wrap around. In academic practice, you should not assume overflow behavior is safe. Instead, you should check ranges, use larger types when needed, or use libraries for big integers when exact very large values are required.

Fig. 2

Data types range from simple to complex, impacting performance and safety.



Data types move from simple to complex. Basic numbers, characters, and Booleans are handled directly by hardware. Arrays, pointers, and references build on basics. Structures, classes, and records model real objects, improving organization while affecting performance, memory use, and program safety.

3.2.2 Fixed-Point As A Special Case

Some systems use fixed-point arithmetic, especially in finance or embedded systems. Fixed-point keeps a fixed number of decimal places, such as cents for money. This can reduce rounding problems compared to floating-point. However, fixed-point still has range limits and needs careful scaling.

3.3 Typing Systems

Typing systems describe when and how a language checks types. This affects program safety, developer workflow, and error detection. Understanding the type system helps you explain why an error appears at compile time in one language but at run time in another.

3.3.1 Static Typing

In static typing, the compiler checks type rules before execution. This can prevent many bugs, such as mixing incompatible types. Static typing can also support optimization because the compiler knows what operations to use. However, static typing may require more explicit declarations and can feel strict to beginners.

3.3.2 Dynamic Typing

In dynamic typing, the program checks types during execution. This can feel flexible, especially for quick prototypes. But it can also allow type errors to appear later, when a path of code is finally executed. In academic work, it is helpful to state the language typing model because it shapes how you test and validate results.

3.4 Mutability And Immutability

Mutability describes whether a value can change after it is created. Many basic numeric types behave like values, and assignments create new values. Some string models treat strings as immutable, meaning they cannot be changed in place. Immutability can improve safety because values do not change unexpectedly. Mutability can improve efficiency in some tasks, such as building text quickly using a mutable buffer. In academic writing, mutability is worth noting when discussing performance and memory behavior.

4.0 EXAMPLES AND CASE STUDIES

Examples connect type rules to real tasks. In each case, type choice affects correctness, memory use, and program clarity. A good example also states why one type is better than another for the given problem.

4.1 Example: Student Record

Imagine a student record with fields: ID number, name, age, and pass/fail status. The ID and age are best stored as integers because they are whole numbers. The name is stored as a string because it is text. The pass/fail status is stored as a Boolean because it has two states. This design is clear and matches meaning. It also helps prevent errors, such as trying to add a name to a number. In an academic report, you can justify each field type by describing the expected values and their limits.

4.2 Example: Prices And Money

Money often needs exact results. If you store money as floating-point, you may see tiny rounding differences, such as $0.1 + 0.2$ not matching 0.3 exactly. A common solution is to store money in the smallest unit, like cents, using integers. For example, ₹199.50 can be stored as 19950 paise. Then addition and subtraction stay exact within the integer range. If decimals are needed for display, you can format the integer back into rupees and paise. In academic writing, this is a strong example of choosing types based on required accuracy.

4.3 Case Study: Integer Overflow In A Counter

Consider a program that counts page views on a busy website. If the counter type is too small, it may overflow after many visits. If the system uses wraparound behavior, the count can suddenly become negative or restart at zero. This can harm reports and decisions. A safer approach is to use a larger integer type, or a big-integer library if needed. Another approach is to check for near-maximum values and handle the situation. This case study shows that type choice must consider worst-case growth, not only average use.

4.4 Case Study: Floating-Point Surprise In A Grade Average

Imagine a program that averages quiz scores like 8.7, 9.3, and 7.9. Floating-point is reasonable because averages can be fractional. However, the final printed average may show 8.633333333 instead of a neat value. Also, comparing floating-point values using exact equality can fail. A better method is to compare using a tolerance, such as “within 0.01.” This case shows that

floating-point is useful but must be used with correct comparison methods and rounding for display.

4.5 Case Study: Characters, Strings, And Encoding In Names

Consider storing names from many languages. If a system assumes a limited encoding, some characters may appear as question marks or boxes. This is not only a technical problem; it is also a data quality issue. A better approach is to use Unicode-aware strings and to keep encoding consistent when reading and writing files. In academic reporting, you can note encoding as part of the system design, especially when data includes multilingual text.

4.6 Case Study: Boolean Logic In Safety Rules

Imagine a safety system that checks rules such as “temperature is safe” and “door is closed.” Each rule can be a Boolean. The final decision can be computed using AND and OR. For example, the system may allow operation only if all key rules are true. Booleans make this logic clear. Using integers like 0 and 1 can work, but it is less clear and can allow mistakes such as using 2 by accident. This case supports the idea that types improve both correctness and readability.

1. Standard data types are basic building blocks used in many programming languages.
2. A data type gives meaning to bits and controls what operations are allowed.
3. Integers store whole numbers and can overflow if the range is exceeded.
4. Floating-point stores fractional values but can introduce rounding differences.
5. Characters and strings store text, and encoding affects how text is represented.
6. Booleans store true/false values and support clear decision logic.
7. Type conversion can help, but unsafe casting can lose information.
8. Good type choice depends on range, precision, memory cost, and meaning.

5.0 Keyword Touch Vocabulary Meaning

Keyword	Simple Meaning	Touch Vocabulary Sentence
Data type	Rule for storing and using a value	A data type tells the computer how to treat a value.
Integer	Whole number type	An integer is used to count students in a class.
Floating-point	Decimal number type	Floating-point helps store an average score with decimals.
Overflow	Value exceeds type range	Overflow can turn a large count into a wrong number.
Character	Single text symbol	A character can be a letter like A or a symbol like #.
String	Sequence of characters	A string can store a person's full name.
Boolean	True/false type	A Boolean can represent whether a user is logged in.
Encoding	Rule for text storage	Encoding decides how letters are saved as bytes.
Casting	Converting types	Casting can change an integer into a floating-point value.
Precision	Level of detail in decimals	Precision affects how many decimal digits a value keeps.